Multiplicity-Atomic Primes (MAP)

This document formalizes a “prime by multiplicity profile” framework that (i) treats multiplicity as a numerical invariant of local complexity, (ii) interfaces naturally with intersection multiplicity via derived tensor products, and (iii) supports an optional extremal/“curvature–torsion” characterization that distinguishes primes from near-prime regimes (prime powers, squarefree composites, highly composite).

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0. Guiding idea

Instead of taking divisibility as primitive, encode an arithmetic object by its **local multiplicity profile** across “prime points.”

- For integers, this is the assignment $p \mapsto v_p(n)$.
- Geometrically, it is a **0-cycle** (effective divisor) on $\mathrm{Spec}(\mathbb{Z})$ supported on height-1 points.
- Categorically, it is the multiset of Jordan–Hölder multiplicities of simple torsion objects.

This lens generalizes cleanly to Dedekind domains and 1-dimensional regular schemes.

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1. Ambient setting

1.1 Dedekind domain model

Let A be a Dedekind domain with fraction field K.

- Height-1 primes $\mathfrak{p} \in \mathrm{Spec}(A)$ are the “prime points.”
- Localizations $A_\mathfrak{p}$ are DVRs.

For a nonzero ideal $I \subset A$, define the torsion module

$$ M_I := A/I. $$

Integer case: $A = \mathbb{Z}$, $I = (n)$, $M_{(n)} \cong \mathbb{Z}/n\mathbb{Z}$.

Polynomial case: $A = \mathbb{F}_q[t]$, $I = (f)$, $M_{(f)} \cong \mathbb{F}_q[t]/(f)$.

1.2 Sheaf model (1-dimensional regular scheme)

Let X be a 1-dimensional regular scheme.

- Codimension-1 points $x \in X^{(1)}$ play the role of primes.
- Local rings $\mathcal{O}_{X,x}$ are DVRs.

For a torsion coherent sheaf $\mathcal{F}$, localize to $\mathcal{F}_x$.

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2. Multiplicity profile as a local length invariant

2.1 Module/ideal multiplicity profile

For a torsion A-module M of finite length (e.g. $M = A/I$), define the **multiplicity profile**

\$\$ \nu_M(\mathfrak p) := \operatorname{length}{*A*)\in \mathbb N. \$\$}}(M_{\mathfrak p

For $M_I = A/I$, write $\nu_I(\mathfrak{p}) := \nu_{A/I}(\mathfrak{p})$.

Facts (Dedekind domain):

- $\nu_I(\mathfrak{p}) = 0$ for all but finitely many $\mathfrak{p}$.
- If $I = \prod_{\mathfrak{p}} \mathfrak{p}^{a_{\mathfrak{p}}}$ (unique factorization of ideals), then \$\$ \nu_I(\mathfrak p)=a_{\mathfrak p}. \$\$

Integer specialization: $I = (n) \subset \mathbb{Z}$ gives $\nu_{(n)}(p) = v_p(n)$.

2.2 Multiplicity cycle / multiplicity field

Define the **multiplicity cycle** (effective divisor / measure)

\$\$ \mu(M) := \sum_{\mathfrak p\in\mathrm{Spec}(A)^{(1)}} \nu_M(\mathfrak p)\,[\mathfrak p]. \$\$

In the sheaf model:

\$\$ \mu(\mathcal F) := \sum_{x\in X^{(1)}} \operatorname{length}{*\mathcal O*(\mathcal F_x)\,[x]. \$\$}

Interpretation: μ is the canonical "multiplicity field" assigning mass ν to each prime point.

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3. Multiplicity-Atomic Primes (MAP)

3.1 Definition (MAP object)

A torsion module M (finite length) over a Dedekind domain A, or a torsion coherent sheaf $\mathcal{F}$ on a 1-dimensional regular scheme X, is **Multiplicity-Atomic Prime (MAP)** if its multiplicity cycle is a single unit spike:

- **Single support:** $\mu(M) = [\mathfrak{p}]$ for some height-1 prime $\mathfrak{p}$ (equivalently $\nu_M(\mathfrak{q}) = 0$ for $\mathfrak{q}\ /\ \mathfrak{p}$)=
- **Unit thickness:** $\nu_M(\mathfrak{p}) = 1$.

Equivalently, in the Dedekind domain case:

$$ M\cong A/\mathfrak p. $$

Integer case: MAP objects are exactly $\mathbb{Z}/p\mathbb{Z}$, i.e. primes.

3.2 Near-MAP strata (spectrum)

The multiplicity cycle yields a natural stratification:

- **Prime powers:** $\mu = k[\mathfrak{p}]$ (single support, thickness k).
- **Squarefree composites:** $\mu = \sum_{i=1}^{r}[\mathfrak{p}_i]$ with $r \geq 2$.
- **Highly composite:** many $\mathfrak{p}$ with larger ν.

This gives a spectrum of "prime-likeness" measured by dispersion of μ and by thickness.

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4. Derived intersections as multiplicity overlap

4.1 Derived tensor product as intersection

For torsion A-modules M, N define their **derived intersection complex**

$$ M\otimes^{\mathbf L}_A N $$

with homology $\mathrm{Tor}_i^A(M, N)$.

Localizing at $\mathfrak{p}$ reduces to the DVR $A_{\mathfrak{p}}$, so the overlap is controlled prime-by-prime.

4.2 Tor-min rule (local DVR calculation)

Let R be a DVR with uniformizer π. For $a, b \geq 1$, set $M_a = R/(\pi^a)$, $N_b = R/(\pi^b)$. Then:

$$ M_a\otimes_R N_b\cong R/(\pi^{\min(a,b)}),\qquad \mathrm{Tor}_1^R(M_a,N_b)\cong R/(\pi^{\min(a,b)}),\qquad \mathrm{Tor}_i^R(M_a,N_b)=0\ (i>1). $$

Consequences:

- Local overlap thickness is $\min(a, b)$.
- Both Tor_0 and Tor_1 carry the same local length.

4.3 Derived minimal-overlap characterization of MAP

Let M be MAP supported at $\mathfrak{p}$.

- If N has disjoint support ($\nu_N(\mathfrak{p}) = 0$), then $(M \otimes^{\mathbf{L}}_A N)_{\mathfrak{p}} \simeq 0$, and in fact the entire derived tensor product vanishes after localization at every prime.
- If N is supported at $\mathfrak{p}$, then local overlap is exactly $\min(1, \nu_N(\mathfrak{p})) = 1$ in Tor_0 and Tor_1.

This implements the slogan: **MAP objects have minimal nontrivial derived intersection, occurring only at the same prime point, and only with unit thickness.**

> Note: In 1-dimensional settings, Serre's alternating Euler characteristic $\sum(-1)^i \ell(\mathrm{Tor}_i)$ often cancels because Tor_0 and Tor_1 have equal length. Here we use **Tor-length profiles** (or $\ell(\mathrm{Tor}_0)$ alone) as the multiplicity signal.

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5. Extremal functionals ("curvature / torsion") on multiplicity profiles

This section builds optional, computable functionals on μ that distinguish:

- localization across supports (how many spikes),
- thickness (prime powers vs unit thickness),
- overall dispersion (highly composite).

5.1 Normalized distribution on primes

For an integer $n \geq 2$, define:

$$\Omega(n)=\sum_p v_p(n),\qquad \omega(n)=\#\{p:\ v_p(n)>0\},\qquad \pi_n(p)=\frac{v_p(n)}{\Omega(n)}.$$

Analogously over Dedekind domains, normalize ν by total mass.

5.2 Dispersion (entropy)

Define the entropy

$$H(n)= -\sum_{p:\ v_p(n)>0} \pi_n(p)\log \pi_n(p).$$

Properties:

- $H(n) = 0$ iff μ_n has single support (i.e. n is a prime power).
- For squarefree composites with r distinct primes, $H(n) = \log r$.

5.3 Thickness penalty (torsion-thickness)

Define

$$T(n) := \Omega(n)-\omega(n)=\sum_p \max\{v_p(n)-1,0\}.$$

Properties:

- $T(n)=0$ iff n is squarefree.
- Prime powers p^k satisfy $T(p^k)=k-1$.

5.4 MAP score

For $\lambda>0$ define

$$ J(n):=H(n)+\lambda\,T(n). $$

Then:

- $J(n)=0$ iff n is prime.
- Prime powers: $H=0, T>0$.
- Squarefree composites: $T=0$, $H>0$.
- Highly composite: typically both H and T large.

This gives a canonical, non-tautological “prime-likeness” scalar that separates the two main failure modes (multi-support vs thickness).

5.5 Optional filtration curvature (order-based)

Define the cutoff filtration

$$ F_n(x):=\sum_{p\le x} v_p(n) $$

(step function in x). One can define a discrete curvature energy via second differences in x (choice-dependent, but computable).

Caution: order-based curvature depends on the chosen filtration (e.g., by $p\le x$, by $\log p$, by Chebyshev weights). Robustness should be tested by comparing several natural filtrations.

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6. Theorems and core propositions (statements)

Theorem 6.1 (Classification of MAP modules over Dedekind domains)

Let A be a Dedekind domain and M a finite-length torsion A-module. Then M is MAP iff $M\cong A/\mathfrak{p}$ for some height-1 prime $\mathfrak{p}$.

Proposition 6.2 (Local derived overlap)

Let $M=A/I$, $N=A/J$. For each height-1 prime $\mathfrak{p}$, localizing at $\mathfrak{p}$ gives:

$$ \ell_{A_{\mathfrak p}}\big(\mathrm{Tor}0^A(M,N)\big) = \ell_{A_{\mathfrak p}}\big(\mathrm{Tor}1^A(M,N)\big) = \min{\nu_I(\mathfrak p),\nu_J(\mathfrak p)}. $$

and $\mathrm{Tor}_i = 0$ for $i > 1$ after localization at a height-1 prime.

Corollary 6.3 (Minimal nontrivial overlap for MAP)

If M is MAP supported at $\mathfrak{p}$, then for any N:

- If $\nu_N(\mathfrak{p}) = 0$, then $(M \otimes^{\mathbf{L}}_A N)_{\mathfrak{p}} \simeq 0$.
- If $\nu_N(\mathfrak{p}) > 0$, then the local derived overlap thickness equals 1.

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7. Predictions / expected structural outcomes

1. **Spectral stratification:** The pair $(H(n), T(n))$ partitions integers into regimes:

2. primes $(0, 0)$,

3. prime powers $(0, > 0)$,
4. squarefree composites $(> 0, 0)$,

5. mixed composites $(> 0, > 0)$.

6. **Derived overlap detects gcd-loci locally:** The pointwise overlap $\min(v_p(m), v_p(n))$ is exactly the local derived thickness; the full Tor-length profile can be read as “hidden multiplicity.”

7. **Generalization robustness:** Over $\mathbb{F}_q[t]$, the same (H, T) and MAP definitions pick out irreducibles as MAP, prime powers as thickness-only deviations, etc.

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8. Fast validation pathway

8.1 Integer sandbox

- Compute $v_p(n)$, $\Omega(n)$, $\omega(n)$, $H(n)$, $T(n)$, $J(n)$ for $2 \leq n \leq N$.
- Verify $J(n) = 0$ iff n is prime.
- Visualize clusters in (H, T) space.

8.2 Structural (derived) validation

- Prove the DVR Tor-min rule.
- Deduce Proposition 6.2 by localization.

8.3 Generalization test

- Repeat computations for $A = \mathbb{F}_q[t]$ up to degree d and compare distributions of (H, T).

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9. Notes and cautions

- In dimension 1, the alternating Serre intersection multiplicity $\sum(-1)^i \ell(\mathrm{Tor}_i)$ can cancel; here the multiplicity signal is carried by Tor-length profiles (or $\ell(\mathrm{Tor}_0)$ alone).
- Over general rings, “simple module” corresponds to maximal ideals; to keep “prime points” aligned with height-1 primes, work in the divisor/Dedekind/regular-codim-1 setting.
- Order-based curvature requires choosing a filtration; test robustness under several natural filtrations.

10. References

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